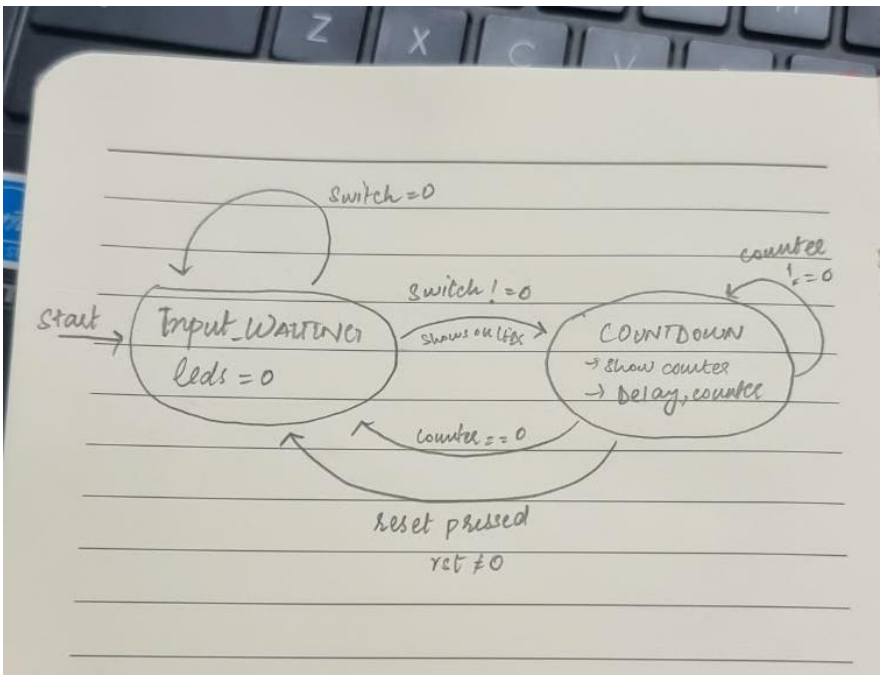
Lab 10 – Worksheet

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Note: Assumptions and logics should be explained separately in tasks after the task results.

Task 1

FSM of assembly implementation of FSM designed in Lab 05



Task 2

Complete assembly source file implementing the FSM.

```
.equ SW_ADDR,     0x100
.equ LED_ADDR,    0x104
.equ RST_ADDR,    0x108
.equ STACK_TOP,   0x1F0
.equ DELAY_SHIFT, 20

main:
      addi sp, x0, STACK_TOP
      addi s0, x0, SW_ADDR
      addi s1, x0, LED_ADDR
      addi s2, x0, RST_ADDR

INPUT_WAITING:                # state 1
      addi s4, x0, 0
      addi s3, x0, 0
      sw   x0, 0(s1)
iw_poll:
      lw   t0, 0(s2)
      bne  t0, x0, INPUT_WAITING
      lw   a0, 0(s0)
      beq  a0, x0, iw_poll

      addi s4, x0, 1
      jal  ra, countdown
      beq  x0, x0, INPUT_WAITING

countdown:
      addi sp, sp, -8
      sw   ra, 4(sp)
      sw   s3, 0(sp)
      addi s3, a0, 0
cd_loop:
      beq  s3, x0, cd_done
      sw   s3, 0(s1)
      jal  ra, delay
      bne  a0, x0, cd_reset
      addi s3, s3, -1
      beq  x0, x0, cd_loop
```

```
cd_done:
        addi a0, x0, 0
        beq  x0, x0, cd_exit
cd_reset:
        addi a0, x0, 1
cd_exit:
        lw   s3, 0(sp)
        lw   ra, 4(sp)
        addi sp, sp, 8
        jalr x0, 0(ra)

delay:
        addi t1, x0, 1
        addi t2, x0, DELAY_SHIFT
        sll  t1, t1, t2
d_loop:
        lw   t0, 0(s2)
        bne  t0, x0, d_reset
        addi t1, t1, -1
        bne  t1, x0, d_loop
        addi a0, x0, 0
        jalr x0, 0(ra)
d_reset:
        addi a0, x0, 1
        jalr x0, 0(ra)
```

Task 3

*Design module code of **instruction memory module** containing memory initialized with assembly instructions:*

```
Instructionmemory:
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10/29/2026 11:07:48 AM
// Design Name:
// Module Name: instructionMemory
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////

// Description: Lab 10 Byte-Addressable Instruction Memory with NOP fallback
//////////////////////////////////////////////////////////////////////////////////

`timescale 1ns / 1ps

module instructionMemory #(
    parameter OPERAND_LENGTH = 31
)(
    input      [OPERAND_LENGTH:0] instAddress,
    output reg [31:0]             instruction
);

    reg [7:0] memory [0:255];
    integer i;

    initial begin
        // Initialize memory with standard RISC-V NOP (addi x0, x0, 0 =
32'h00000013)
        // Little-endian byte order: 13, 00, 00, 00
```

```
        for (i = 0; i < 256; i = i + 4) begin
            memory[i]   = 8'h13;
            memory[i+1] = 8'h00;
            memory[i+2] = 8'h00;
            memory[i+3] = 8'h00;
        end

        // Load assembled machine code from file
        $readmemh("instructions.mem", memory);
    end

    // Little-endian combinational instruction read
    always @(*) begin
        if (instAddress <= 252) begin
            instruction = {memory[instAddress + 3],
                           memory[instAddress + 2],
                           memory[instAddress + 1],
                           memory[instAddress]};
        end else begin
            instruction = 32'h00000013; // Return NOP if address out of range
        end
    end

endmodule
```

memory file

```
// --- main: setup stack & MMIO pointers ---
13 01 00 1f
13 04 00 10
93 04 40 10
13 09 80 10

// --- INPUT_WAITING: clear LEDs & poll ---
13 0a 00 00
93 09 00 00
23 a0 04 00
83 22 09 00
e3 98 02 fe
03 25 04 00
e3 0a 05 fe
13 0a 10 00
ef 00 80 00
e3 0e 00 fc

// --- countdown subroutine ---
13 01 81 ff
23 22 11 00
```

```
23 20 31 01
93 09 05 00
63 8c 09 00
23 a0 34 01
ef 00 c0 02
63 1a 05 00
93 89 f9 ff
e3 06 00 fe
13 05 00 00
63 04 00 00
13 05 10 00
83 29 01 00
83 20 41 00
13 01 81 00
67 80 00 00

// --- delay subroutine ---
13 03 10 00
93 03 40 01
33 13 73 00
83 22 09 00
63 9a 02 00
13 03 f3 ff
e3 1a 03 fe
13 05 00 00
67 80 00 00
13 05 10 00
67 80 00 00
```

Testbench

```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10/29/2026 10:37:44 AM
// Design Name:
// Module Name: instructionMemory_tb
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
```

```
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////

`timescale 1ns / 1ps

module instructionMemory_tb;

   reg  [31:0] pc;
   wire [31:0] instruction;

   instructionMemory #(
      .OPERAND_LENGTH(31)
   ) uut (
      .instAddress (pc),
      .instruction (instruction)
   );

   integer k;

   initial begin
      #10;
      $display("     VERIFYING INSTRUCTION MEMORY (LAB 10 TASK 3)
");

      // Fetch instructions 0 to 41 (Byte addresses 0 to 164)
      for (k = 0; k < 42; k = k + 1) begin
         pc = k * 4;
         #10;
         $display("PC = 0x%03h (%03d) | Instruction = 0x%08h", pc, pc,
instruction);
      end

      // Test past program space for default NOP padding
      pc = 180;
      #10;
      $display("-----------------------------------------------------------");
      if (instruction === 32'h00000013) begin
         $display("PC = 0x%03h (%03d) | Instruction = 0x%08h [PASS: NOP
verified]", pc, pc, instruction);
      end else begin
         $display("PC = 0x%03h (%03d) | Instruction = 0x%08h [FAIL: Expected
00000013]", pc, pc, instruction);
      end

$display("==================================================
==========");
```

```
        $finish;
    end

endmodule
```

Lab 10: State-Based Control Flow Using RISC-V Assembly

Points Distribution

Task No.	LR2 Code	LR 5 Results	LR11 Design	AR7
Task 1	-	-	/20	/20
Task 2	/20	/20		
Task 3	/20	-		
Total Points	/80			/20
CLO Mapped	CLO 1			

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR2	**Program/Code / Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
AR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.